Project Plan

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Comparison of Direct and Cascaded High-Resolution Spatial Downscaling of Wildfire Data

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Abstract

Wildfires are destructive and a rapidly evolving hazard. With climate change, they become increasingly severe and frequent. This increases the need for a higher-resolution wildfire data for better monitoring and risk assessment. Geostationary satellites (GEO) have high-temporal resolution, but lacks the resolution needed for a detailed wildfire risk assessment. While low orbit satellites (LEO) provide better spatial resolution, but low temporal resolution to monitor rapid wildfire spread. Existing generative downscaling models have demonstrated their effectiveness in producing sharp, high-resolution remote sensing data. But many still suffers from artefacts, blurriness, and hallucinations. This project aims to achieve a high-resolution spatial downscaling of wildfire data using downscaling models that integrates additional physical covariates, such as topography and landcover to mitigate physically implausible outputs. The project compares direct (2km to 30m) and cascaded (2km to 375m to 30m) downscaling pipelines and their effectiveness in producing high-fidelity, high-resolution wildfire data. The project compares diffusion (EDiffSR), attention U-Net (HAUNet), and transformer-based (ESTNet) downscaling models.

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1 Problem Description and Motivation

In recent years, due to rising popularity in Net-Zero initiatives, carbon investors are increasingly funding carbon reduction and removal projects, such as reforestation (Lambert and Turner, 2024). However, these carbon projects are at risk because wildfires can destroy and release the carbon back into the atmosphere. With climate change, wildfires are becoming increasingly severe and frequent (Jones et al., 2020). Which increases the need for better wildfire monitoring and prediction for risk assessment. This means high-spatial and high-temporal resolution wildfire data is highly desired. However, high-temporal resolution data obtained from geostationary satellites (GEO) have limited spatial resolution, which makes them difficult to use for accurate risk assessment. While low orbit satellites (LEO) offers high-spatial resolution, but low-temporal resolution.

To bridge this gap, existing methods uses Deep Learning (DL) models for downscaling due to their ability in recognising non-linear spatiotemporal patterns. However, the resolution gap between GEO and LEO is significant (2km - 30m). This increases the risk of producing physically implausible or unrealistic outputs, as a single low-resolution (LR) pixel can correspond to different high-resolution (HR) details (Z. Wang et al., 2020). In this project, to mitigate this limitation, the model integrates additional physical information, such as HR topography and landcover to help guide the downscaling process. To further improve downscaling, the project aims to investigate whether cascaded downscaling (2km to 375m to 30m) produces more accurate and better results compared to direct downscaling (2km to 30m). By breaking up the downscaling process, the model trains on a smaller resolution gap, aiming to mitigate physically implausible outputs. This project will compare the cascaded and direct pipeline performance of different downscaling architectures, including hybrid attention U-Net, a transformer-based model, and a diffusion-based model.

2 Literature Review

2.1 Current State of Remote Sensing Downscaling

Traditional downscaling methods often fail to capture the complex non-linear patterns and variability in remote sensing data. In recent years, DL methods have shown promising results that outpaces traditional approaches (Sdraka et al., 2022). In particular, Convolutional Neural Networks (CNNs) and U-Net architectures have been widely used for their ability to capture spatial features. Generative Adversarial Networks (GANs) have also been used for their ability to generate sharp images. Recently, Transformer and Diffusion-based approaches have been proposed in super-resolution research. Transformers use attention mechanisms to capture long-range dependencies (J. Wang et al., 2025), while Diffusion models shows good generative ability and fidelity (Y. Wang et al., 2023).

However, many of these approaches have limitations, particularly in preserving fine-grained textures and geometric structures (Qi et al., 2026). This is problematic for wildfire data, where the spatial patterns and textures are crucial for accurate risk assessment. To address these limitations, there is a need for domain-specific architectures that can better capture the characteristics of wildfire data.

In the current literature, there is also a lack of studies that compares the performance of direct and cascaded downscaling pipelines. Some studies have explored cascaded approaches, such as AMCN-Geoi-DBN from Jing et al., 2022, and CLIT from Chen et al., 2023, but more research is needed to fully understand the benefits and limitations in the context of wildfires. Cascaded downscaling could also introduce additional complexity and computational cost. Therefore more research is needed to also investigate whether there are significant benefits to cascaded downscaling.

2.2 Selected Downscaling Architectures for Comparison

2.2.1 HAUNet

Among CNN-based approaches, HAUNet (J. Wang et al., 2023) uses a hybrid attention mechanism to fuse multiscale, local and global features. This allows it to capture both fine-grained details and broader contextual information. HAUNet will be selected as one of the architectures for comparison. Because while U-Nets capture spatial features effectively, combining them with attention mechanism makes it a strong baseline for capturing spatially heterogeneous patterns in wildfire data.

2.2.2 ESTNet

ESTNet (Kang et al., 2024) is a transformer-based architecture that uses efficient spatial-temporal attention to capture long-range dependencies in the data. It is based on Swin-Transformers with group attention. It uses multi-window self-attention for better efficiency without sacrificing reconstruction quality. For transformer-based approach, ESTNet will be selected for comparison due to their ability in capturing long-range dependencies, making them a suitable candidate for downscaling wildfire data, which has complex spatial patterns. ESTNet specifically is preferred over alternatives such as SwinIR (Liang et al., 2021) because their efficiency makes them more suitable for the large amount of data and inputs used in this project.

2.2.3 EDiffSR

EDiffSR (Xiao et al., 2023) is a conditional diffusion-based model. It uses simplified channel attention for better noise prediction quality, and Conditional Prior Enhancement Module (CPEM) to extract the most valuable information from the LR input. EDiffSR is selected as the diffusion-based model for comparison because it effectively leverages prior information and promotes the condition with more informative input.

3 Methodology

The architectures (HAUNet, ESTNet, EDiffSR) for comparison is tentatively selected based on their performance in super-resolution tasks, and their suitability for the specific requirements of the wildfire downscaling. The reasons for selecting each architecture are outlined in 2.2. However, the final selection of architectures and datasets may be adjusted as the project progresses.

3.1 Datasets

3.1.1 Input Datasets

The input data are:

- GOES-17 channel 7 (IR shortwave) of Level 1B (L1B) Radiances product (ABI-L1B-Rad). Has spatial resolution of 2km at every 5 mins. This channel is selected because it is commonly used to identify hot regions (Badhan et al., 2024). This will be the main coarse input.
- Copernicus DEM GLO-30: Global 30m Digital Elevation Model with 30m resolution (European Space Agency and Airbus, 2022). DEM band will be used.
- ESA WorldCover 10m v100 landcover dataset with 10m resolution. This dataset have 11 land cover classes, including tree cover, shrubland, grassland, and permanent water bodies (Zanaga et al., 2021).

- Other potential covariates include weather data, such as wind speed and direction, temperature, and humidity, which can be obtained from reanalysis or operational datasets. The inclusion of these covariates will depend on their availability, relevance and time constraints.

3.1.2 Target Datasets

The target data are:

- VIIRS (S-NPP) Band Active Fire with 375 m spatial resolution, VNP14IMGTDL_NR (Earth Science Data Systems, 2025) will be used as the target data for the 2km to 375m stage downscaling. The I-4 channel brightness temperature (3.55-3.93 µm) band will be used.
- Landsat-8 active fire dataset includes data for fire scenes taken from August and September 2020 with 30m resolution (de Almeida Pereira et al., 2021). This will be the target data for the final 375m to 30m stage downscaling. Fire-sensitive SWIR and thermal bands will be used.

3.2 Experimental Design

The project will evaluate the performance of the selected architectures in both direct and cascaded downscaling pipelines. The direct pipeline will involve training the model to directly downscale from 2km to 30m resolution. The cascaded pipeline will involve two stages: the first stage downscales from 2km to 375m (trained on GOES and VIIRS data), and the second stage will further downscale from 375m to 30m (trained on VIIRS and Landsat-8 data). The final 30m resolution output from both pipelines will be compared against the Landsat-8 active fire dataset for evaluation.

Table 1: Summary of Experimental Configurations

Experiment	Architecture	Pipeline	Input Resolution	Output Resolution	Model Type
E1	HAUNet	Direct	2km (GOES-17)	30m (Landsat-8)	Hybrid Attention U-Net
E2	HAUNet	Cascaded	Stage 1: 2km → 375m Stage 2: 375m → 30m	30m (Landsat-8)	Hybrid Attention U-Net
E3	EDiffSR	Direct	2km (GOES-17)	30m (Landsat-8)	Diffusion
E4	EDiffSR	Cascaded	Stage 1: 2km → 375m Stage 2: 375m → 30m	30m (Landsat-8)	Diffusion
E5	ESTNet	Direct	2km (GOES-17)	30m (Landsat-8)	Transformer
E6	ESTNet	Cascaded	Stage 1: 2km → 375m Stage 2: 375m → 30m	30m (Landsat-8)	Transformer

3.3 Evaluation Metrics

The metrics for evaluation will include:

- Fréchet Inception Distance (FID): Measures the distance between the distribution of generated images and real images. Lower FID indicates better alignment.
- Peak Signal-to-Noise Ratio (PSNR): Measures how well the reconstructed image preserves the image quality. Higher PSNR indicates better quality.
- Structural Similarity Index (SSIM): Measures the similarity between two images based on their structural information. Higher SSIM indicates better quality.

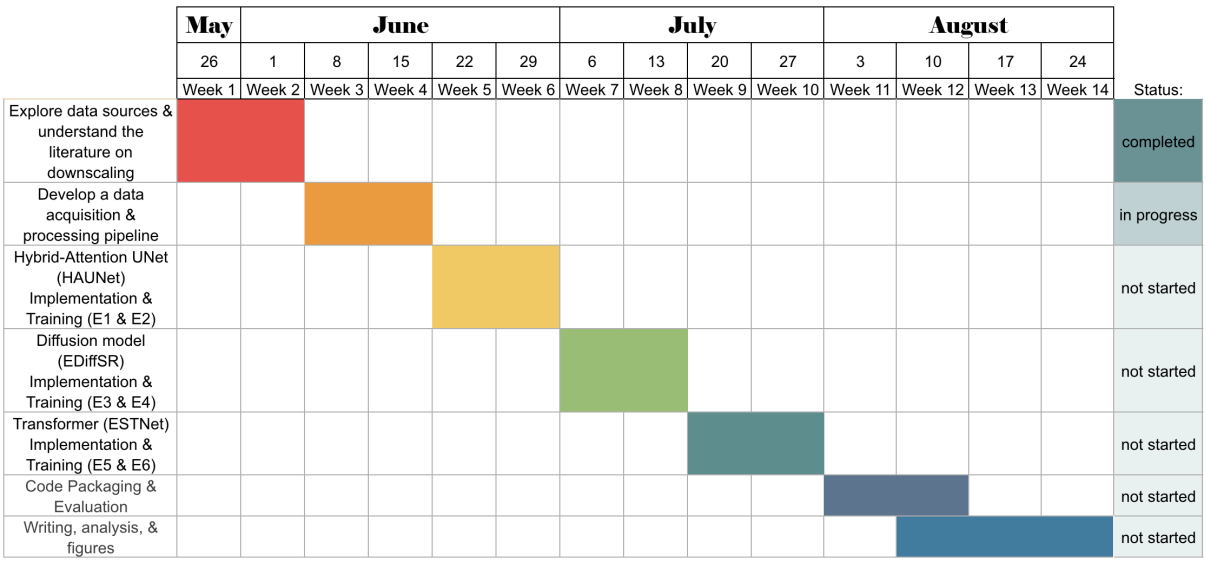


Figure 1: Gantt Chart of Project Timeline and Milestones

- Coarse-Grid to Fine-Grid Mean Squared Error (CG-FG MSE): Measures the MSE between the coarse-grid input and the target data. This quantifies the baseline error generated by simply interpolating lower-resolution data.
- Super-Resolution to Fine-Grid Mean Squared Error (SR-FG MSE): Measures the MSE between the downscaled output and the target data. Lower MSE indicates better performance.

4 Progress to Date

The project is currently in the early stages of data collection and preprocessing. The data acquisition and processing pipeline is being developed. The key milestones for the project is shown in Fig. 1.

5 Future Plans

The next steps for the project include:

1. Developing the data preprocessing pipeline to prepare the input and target datasets for training. This includes aligning the datasets spatially and temporally, and integrating the additional covariates.
2. Implementing the selected architectures and training them on the respective datasets for both direct and cascaded pipelines.
3. Evaluating the performance of the models using the defined metrics and comparing the results between direct and cascaded pipelines.
4. Conducting a detailed analysis of the results to understand the strengths and limitations of each approach, and identifying potential areas for improvement.
5. Preparing the final report and presentation summarising the findings and insights from the project.

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